

09/11/2026: Lists

CSCI 246: Discrete Structures

Textbook reference: Sec. 8, Scheinerman

Announcements

- Early-access version of the slides should be posted to the course repo before class starts. (The early-access version will not yet contain the group exercises or, if relevant, a quiz).
- The full version of the slides are posted to the course repo after class ends. **Starting today, the full slide deck will be posted to the repo automatically at 3pm.**

Today's Agenda

- Quiz (≈ 15 mins)
- Review of Boolean Algebra (≈ 10 mins)
- Mini-lecture: Lists (≈ 10 mins)
- Group exercises (≈ 10 mins)

When choosing 1 burger, 1 side serve and 1 drink ...
how many combinations are possible?

burgers



- beef
- chicken
- fish
- vegetarian

4 possibilities

side serves



- fries
- salad

2 possibilities

drinks



- juice
- coffee
- cola
- water

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Solution. There are $4 \times 2 \times 4 = 32$ possibilities.

Definition

A **list** is an ordered sequence of objects.

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Example: All possible meal combinations for the fast food problem

(beef, fries, juice)	(chicken, fries, juice)	(fish, fries, juice)	(veg, fries, juice)
(beef, fries, coffee)	(chicken, fries, coffee)	(fish, fries, coffee)	(veg, fries, coffee)
(beef, fries, cola)	(chicken, fries, cola)	(fish, fries, cola)	(veg, fries, cola)
(beef, fries, water)	(chicken, fries, water)	(fish, fries, water)	(veg, fries, water)
(beef, salad, juice)	(chicken, salad, juice)	(fish, salad, juice)	(veg, salad, juice)
(beef, salad, coffee)	(chicken, salad, coffee)	(fish, salad, coffee)	(veg, salad, coffee)
(beef, salad, cola)	(chicken, salad, cola)	(fish, salad, cola)	(veg, salad, cola)
(beef, salad, water)	(chicken, salad, water)	(fish, salad, water)	(veg, salad, water)

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(beef, fries, water)	(chicken, fries, water)	(fish, fries, water)	(veg, fries, water)
(beef, salad, juice)	(chicken, salad, juice)	(fish, salad, juice)	(veg, salad, juice)
(beef, salad, coffee)	(chicken, salad, coffee)	(fish, salad, coffee)	(veg, salad, coffee)
(beef, salad, cola)	(chicken, salad, cola)	(fish, salad, cola)	(veg, salad, cola)
(beef, salad, water)	(chicken, salad, water)	(fish, salad, water)	(veg, salad, water)

Remark. The sequence is **ordered**: the first entry must be a burger, the second must be a side, and the third must be a drink.

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Poll. What mathematical principle is at play here?

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Multiplication Principle (Theorem 8.2 from Scheinerman)

Consider the two-element lists for which there are n choices for the first element, and for each choice of the first element there are m choices for the second element. Then the number of such lists is nm .

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Solution. We can apply the multiplication principle **iteratively**. So, for example, if we want to form three-element lists for which there are n, m , and k choices for the 1st, 2nd, and 3rd elements respectively, then there are nmk such lists.

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Solution. The number of possible sequences if we draw $k = 6$ balls from the jar of $n = 10$ balls and record the sequence in which they appear is:

$$= \begin{cases} 10^6, & \text{if repetitions are permitted} \\ (10)_6 = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5, & \text{if repetitions are forbidden} \end{cases}$$

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Theorem 8.6 from Scheinerman

The number of lists of length k whose elements are chosen from a pool of n elements

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Notation. The expression

$$(n)_k := \underbrace{n \times (n-1) \times \cdots \times (n-k+1)}_{k \text{ terms}}$$

is called a **falling factorial** and is read as “ n falling k ”.

Practice Reading Quiz

Practice Reading Quiz (Lists)

Five students in our class are randomly selected to form an emergency rock band. The rock band will have a singer, a guitarist, a bassist, a drummer, and a keyboardist. How many different rock bands can be formed in this way?

Notes: (1) Our class has 39 students. (2) If different students play different instruments, it counts as a "different band".

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1. Suppose repetitions are permitted. That is, one student can play multiple instruments. Then $N = 39^5$.
2. Suppose repetitions are forbidden. That is, no student plays more than one instrument (singing counts as an instrument). Then $N = (39)_5 = 39 \times 38 \times 37 \times 36 \times 35$.

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The skill: does order matter, and are repeats allowed?